

RMO(INDIA) – 2013(3)

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
 - All Questions carry equal marks.
 - Answer to each questions starts on a new page. Clearly indicate the question number.
1. Find the number of eight digit numbers the sum of whose digits is 4.
 2. Find all 4-tuples (a, b, c, d) of natural numbers with $a \leq b \leq c$ and $a!+b!+c!=3^d$.
 3. In an acute angled triangle ABC with $AB < AC$, the circle Γ touches AB at B and passes through C intersecting AC again D. Prove that the orthocenter of triangle ABD lies on Γ if and only if and only if it lies on the perpendicular bisector of BC.
 4. A polynomial is called a Fermat polynomial if it can be written as the sum of the squares of two polynomials with integer coefficients. Suppose that $f(x)$ is a Fermat polynomial such that $f(0)=1000$. Prove that $f(x)+2x$ is not Fermat polynomial.
 5. Let ABC be a triangle which is not right angled. Define a sequence of triangles $A_i B_i C_i$, with $i \geq 0$, as follows: $A_0 B_0 C_0$ is the triangle ABC; and for $i \geq 0$, $A_{i+1}, B_{i+1}, C_{i+1}$ are the reflections of the orthocenter of triangle $A_i B_i C_i$ in the sides $A_i B_i, C_i A_i, A_i B_i$ respectively. Assume that $\angle A_m = \angle A_n$ for some distinct natural numbers m, n. Prove that $\angle A = 60^\circ$.
 6. Let $n \geq 4$ be a natural number. Let $A_1 A_2 \dots A_n$ be a regular polygon and $X = \{1, 2, \dots, n\}$. A subset $\{i_1, i_2, \dots, i_k\}$ of $k \geq 3$ and $i_1 < i_2 < \dots < i_k$, is called a good subset if the angles of the polygon $A_{i_1}, A_{i_2}, \dots, A_{i_k}$, when arranged in the increasing order, are in an arithmetic progression. If n is a prime, show that a proper good subset of X contains exactly four elements.

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